

Example: PeV Neutrino in the Earth

Consider UHE neutrino with energy 1 PeV. How far it can travel without interactions?

1. Cross-section on nucleons:

$$\sigma(\nu N) \approx 10^{-33} \text{ cm}^2$$

$$\text{PeV} = 10^{15} \text{ eV}$$

2. Number density of targets:

$$n_N = \rho \cdot N_A = 5.5 \frac{\text{g}}{\text{cm}^3} \cdot \frac{6.02 \cdot 10^{23}}{\text{mole}} \approx \frac{3.3 \cdot 10^{24}}{\text{cm}^3}$$

3. Interaction length:

$$\lambda = \frac{1}{n_N \cdot \sigma} = \frac{\text{cm}^3}{3.3 \cdot 10^{24}} \cdot \frac{1}{10^{-33} \text{ cm}^2} \approx 3 \cdot 10^8 \text{ cm} \approx 3000 \text{ km}$$

Parity